Lösung: Drehscheiben

Scheibe ≙ Hohlzylinder

a) Massen der 2 Scheiben

$$M_A = \rho V_A = \rho \pi \cdot h \left(R_{a,A}^2 - R_i^2\right)$$
$$M_B = \rho V_B = \rho \pi \cdot h \left(R_{a,B}^2 - R_i^2\right)$$

Trägheitsmoment der 2 Scheiben

$$J_A = \frac{1}{2} M_A \cdot \left(R_{a,A}^2 + R_i^2\right) = \frac{1}{2} \rho \pi \cdot h \left(R_{a,A}^4 - R_i^4\right)$$
$$J_B = \frac{1}{2} M_B \cdot \left(R_{a,B}^2 + R_i^2\right) = \frac{1}{2} \rho \pi \cdot h \left(R_{a,B}^4 - R_i^4\right)$$

Gesamtträgheitsmoment der 2 Scheiben

$$J_{ges} = J_A + J_B = \frac{1}{2} \rho \pi \cdot h \left(R_{a,A}^4 - R_i^4\right) + \frac{1}{2} \rho \pi \cdot h \left(R_{a,B}^4 - R_i^4\right)$$
$$= \frac{1}{2} \rho \pi \cdot h \left(R_{a,A}^4 + R_{a,B}^4 - 2R_i^4\right)$$

Beim Auseinandertreffen der Scheiben (kein Drehmoment von außen) gilt Drehimpulserhaltung.

"vorher" "nachher"

$$\vec{L}_A + \vec{L}_B = \vec{L}_{ges}$$
$$L_A + L_B = L_{ges}$$

... , da Drehimpulse parallel

$$J_A \omega_0 + J_B \cdot 0 s^{-1} = J_{ges} \cdot \omega_1$$

$$\omega_1 = \frac{J_A}{J_{ges}} \cdot \omega_0 = \frac{\frac{1}{2} \rho \pi \cdot h \left(R_{a,A}^4 - R_i^4\right)}{\frac{1}{2} \rho \pi \cdot h \left(R_{a,A}^4 + R_{a,B}^4 - 2R_i^4\right)}$$

$$= \frac{R_{a,A}^4 - R_i^4}{R_{a,A}^4 + R_{a,B}^4 - 2R_i^4} \omega_0$$

$$\omega_1 = \frac{(0{,}3\,m)^4 - (0{,}05\,m)^4}{(0{,}3\,m)^4 + (0{,}2\,m)^4 - 2(0{,}05\,m)^4} \, 1 s^{-1} \Rightarrow \boxed{\omega_1 \approx 0{,}84\, s^{-1}}$$

b) $\Delta Q = E^{\text{vorher}} - E^{\text{nachher}}$

$= \frac{1}{2} J_A \cdot \omega_0^2 - \frac{1}{2} J_{ges} \cdot \omega_1^2$

$= \frac{1}{2} J_A \cdot \omega_0^2 - \frac{1}{2} J_{ges} \cdot \left(\frac{J_A}{J_{ges}} \cdot \omega_0 \right)^2$

$= \frac{1}{2} J_A \omega_0^2 \left(1 - \frac{J_A}{J_{ges}} \right)$

$$P = \frac{\Delta Q}{E^{\text{vorher}}} = \frac{\frac{1}{2} J_A \omega_0^2 \left(1 - \frac{J_A}{J_{ges}}\right)}{\frac{1}{2} J_A \omega_0^2} = 1 - J_A / J_{ges}$$

$$= 1 - \frac{J_A}{J_A + J_B} = \frac{(J_A + J_B) - J_A}{J_A + J_B} = \frac{J_B}{J_{ges}}$$

$$= \frac{\frac{1}{2} 8\pi \cdot h \left(R_{a,B}^4 - R_i^4\right)}{\frac{1}{2} 8\pi \cdot h \left(R_{a,A}^4 + R_{a,B}^4 - 2R_i^4\right)} = \frac{R_{a,B}^4 - R_i^4}{R_{a,A}^4 + R_{a,B}^4 - 2R_i^4}$$

$$= \frac{(0{,}2\,\text{m})^4 - (0{,}05\,\text{m})^4}{(0{,}3\,\text{m})^4 + (0{,}2\,\text{m})^4 - 2\,(0{,}05\,\text{m})^4} \approx 0{,}1645 \approx \underline{\underline{16{,}5\,\%}}$$